Veer Test Plan

Course: Computer Science Senior Design I
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Document version: 1.0 | November 11, 2026
Traces to: Requirements Document v1.0 and Technical Specification v1.0 §6 (traceability matrix)

Version and Change Log

Version	Date	Change	Author
1.0	2026-11-11	First test plan. Uses the TC-01 to TC-32 IDs reserved in the traceability matrix. Where a requirement has more than one case, cases are numbered TC-##.1, TC-##.2, TC-##.3 under the same matrix ID, so no matrix row changes.	Team

1. Scope

In scope. Every Must and Should requirement in Requirements Document v1.0, across C1 to C4 and the glue components (auth, notifications, configuration, logging). Must requirements are the major requirements and get 3 cases each.

Component	Owner	Requirements tested
C1 Mobile Navigation	Grady Algire	REQ-03, REQ-04, REQ-15, NFR-02, NFR-07
C1 mic, mute, notifications	Aiden Ward	REQ-10, REQ-11, REQ-12, REQ-13, NFR-09
C2 Speech Understanding	Brady Cooper	REQ-02, REQ-06, NFR-01, NFR-11
C3 Route Planning (incl. weather)	Briar Elliot, Dominic Rowland	REQ-01, REQ-05, REQ-07, REQ-08, NFR-10
C4 Usage Monitor, auth, logging, config	Aiden Ward	REQ-14, NFR-03, NFR-04, NFR-05, NFR-06, NFR-08
Constraints	Grady Algire, Brady Cooper	CON-01 to CON-06 (inspection)

Out of scope.

Item	Reason
REQ-09 avoid highways and tolls (TC-09)	Could priority with no user story yet (Q-02). Deferred to after M-23.
All Won't items in Requirements §7 (accounts, web planner, iOS, offline maps, sponsored ranking, custom hardware, weather history, cloud mute, microservices)	Not being built this release.
Google Maps route quality, OpenAI transcription internals, Open-Meteo forecast accuracy, Android OS mic and TTS, Render uptime	Third-party parts. We test how Veer handles their output and failures, not the services themselves.
EC-16 behavior after the 500-event queue fills	Known limitation, design open (Q-05). We test up to 500 queued events only.
Real monthly cap amounts	Not set yet (G-07). Tests use a $10 test cap.

2. Verification and Validation Strategy

Verification (built right, checked against the spec by the team): every REQ and NFR in scope, through the test cases in §4. **Validation** (right product, checked against stakeholder needs): user stories US-01, US-02, US-04, and US-05 by 3 drivers from outside the team in a parked car at M-20 and on supervised drives at M-23, and US-03 by our advisor reviewing the usage report. Each validator does the story's task with no help and rates it 1 to 5.

Layer	Approx. count	Coverage focus	Tool
Unit	80	Allow-list check, filter-then-rank, C1 state machine, quota math, config schema, redaction	Jest (backend), JUnit (Kotlin)
Integration	30	I2, I4, I7 to I9, auth endpoints with E2 to E4 mocked	Supertest, recorded fixtures
System	50	The cases in §4, run on emulator + staging	Manual + scripts
Acceptance	5	One task per user story, by outside drivers and advisor	Parked car, supervised drive

Speech accuracy (ML). Variables: speaker (3, at least 2 with accents), phrase (10 scripted), cabin noise (engine on, fan on low). Metric: correct constraint rate. Tolerance: 8 of 10 or better per speaker. Trials: 30 (10 phrases × 3 speakers). **Timing.** Every timing case is run 20 times and all 20 must meet the limit, except NFR-10, which is a p95.

Risk order: mute and audio privacy (REQ-11, REQ-12, NFR-01), approval (REQ-03), and keeping the route on failure (NFR-02) are tested first, then budget and data integrity (NFR-04, REQ-14), then the rest.

3. Success Criteria

Area	Metric	Target	How measured	Req ID
Accuracy	Stop detour	≤ 600 s	`extraSeconds` in I4 reply	REQ-01
Usability	Words per spoken reply	≤ 15	Word count of TTS debug log	REQ-02, NFR-07
Reliability	Routes activated without "okay"	0 of 20 trials	`tripRevision` before and after	REQ-03
Performance	"okay" to route active	≤ 3.0 s	Debug log timestamps	REQ-04
Performance	Request to no-match prompt	≤ 3.0 s	Debug log timestamps	REQ-05
Performance	Speech end to repeat prompt	≤ 2.0 s	Debug log timestamps	REQ-06
Usability	Mic state visible during trip	100% of samples	Screenshot every 30 s for 10 min	REQ-10
Performance	Mute to mic stopped	≤ 1.0 s	`micState` change in debug log	REQ-11

Reliability	Force-stop time; audio after mute	≤ 2.0 s; 0 frames	Debug log; server frame counter	REQ-12
Security	Raw audio files stored	0	File scan of phone, DB dump, logs	NFR-01
Reliability	Route kept on failure	34 of 34 edge cases	Active `routeId` unchanged	NFR-02
Security	Provider keys in APK	0	Decompile and grep	NFR-03
Reliability	Spend over cap	$0.00	`monthly_budget.used` vs. cap	NFR-04
Security	Log lines with private data	0	Grep logs for audio, transcript, GPS, tokens	NFR-05
Reliability	Retention	90 days / 12 months	Rows left after retention job	NFR-08
Performance	p95 latency, 5 trips	I2 ≤ 1.5 s, I4 ≤ 2.5 s	k6 load test	NFR-10
Accuracy	Correct constraint rate	≥ 8 of 10 per speaker	Scripted phrase set	NFR-11
Usability	Validation task success	≥ 4 of 5 tasks, rating ≥ 3	Outside driver sessions	US-01 to US-05

4. Test Cases

Setup S1 (used by most cases): staging backend, Pixel 7 emulator (§5), mock GPS route R1 Cincinnati (39.1031, -84.5120) to Columbus (39.9612, -82.9988), destination entered, trip started, mic listening. Clips A-01 to A-14 are played into the emulator mic (§5). “Debug log” means the C1 debug-build event log.

C3 Route Planning

ID	Description	Steps	Expected Result	Status
TC-01.1	Add a named stop (REQ-01)	1. S1. 2. Play A-01 “I’m hungry for McDonald’s.” 3. Read I4 reply in debug log.	Proposal with `selectedPlace` named McDonald’s and `extraSeconds` ≤ 600.	Not Run
TC-01.2	Add a category stop (REQ-01)	1. S1. 2. Play A-02 “find a gas station.” 3. Read I4 reply.	Place is a gas station; `extraSeconds` ≤ 600.	Not Run
TC-01.3	Edge: only match is too far (REQ-01, EC-09)	1. S1 with Maps fixture F-FAR (nearest McDonald’s adds 900 s). 2. Play A-01.	`no_match`; no proposal; `routeId` unchanged; prompt for another place.	Not Run
TC-05	No match prompt time (REQ-05)	1. S1 with F-FAR. 2. Play A-01, 20 times.	Prompt starts ≤ 3.0 s after request in 20 of 20.	Not Run
TC-07	“No” gives next candidate (REQ-07)	1. TC-01.1 steps 1-2. 2. Play A-04 “no.”	A different place is proposed in one sentence.	Not Run
TC-08	Avoid rain, and warn on stale data (REQ-08)	1. S1 with weather fixture W-RAIN. 2. Play A-05 “avoid the rain.” 3. Repeat with W-STALE.	W-RAIN: chosen route has lowest rain exposure. W-STALE: warning spoken; no claim of avoiding rain.	Not Run
TC-25	Latency under load (NFR-10)	1. Run `k6 run load/veer.js` against staging: 5 virtual trips, 100 I2 and 100 I4 requests each.	p95 I2 ≤ 1.5 s; p95 I4 ≤ 2.5 s.	Not Run

C1 Mobile Navigation

ID	Description	Steps	Expected Result	Status
TC-03.1	“okay” activates route (REQ-03)	1. TC-01.1 steps 1-2. 2. Play A-03 “okay.”	`tripRevision` +1; new route on map.	Not Run
TC-03.2	Edge: silence never approves (REQ-03)	1. TC-01.1 steps 1-2. 2. Wait 30 s silent. 3. Repeat 20 times.	`tripRevision` and `routeId` unchanged in 20 of 20.	Not Run
TC-03.3	Failure: unclear reply (REQ-03)	1. TC-01.1 steps 1-2. 2. Play A-06 “um, maybe.” 3. Repeat 20 times.	Route unchanged in 20 of 20; driver asked again.	Not Run
TC-04	Activation time (REQ-04)	1. TC-03.1, 20 times.	Route active ≤ 3.0 s after “okay” ends in 20 of 20.	Not Run
TC-15.1	Spoken turn-by-turn (REQ-15)	1. S1. 2. Play GPS track R1.gpx at 1x.	Every `RouteStep` instruction is spoken before its turn.	Not Run
TC-15.2	Edge: no destination (REQ-15, EC-03)	1. Open app, skip destination. 2. Tap Start.	Trip does not start; app asks for a destination.	Not Run
TC-15.3	Failure: GPS and network lost (REQ-15, EC-33, EC-34)	1. S1 and play R1.gpx. 2. Set network to offline. 3. Pause GPS for 60 s.	Guidance continues; “No connection” and “GPS signal lost” each said once.	Not Run
TC-17.1	Keep route on C2 failure (NFR-02, EC-06, EC-08)	1. S1 with E2 mock set to timeout, then to 503. 2. Play A-01 for each.	`routeId` unchanged; guidance continues.	Not Run
TC-17.2	Keep route on C3 failure (NFR-02, EC-10 to EC-12)	1. S1 with E3 mock set to timeout, then 503, then stale `tripRevision`. 2. Play A-01 for each.	`routeId` unchanged; stale reply ignored.	Not Run
TC-17.3	Full edge-case suite (NFR-02, EC-01 to EC-34)	1. Run `npm run test:edge` on staging plus the C1 cases in `edge/README.md`.	34 of 34 pass, each logged by EC ID.	Not Run
TC-22.1	Reply length (NFR-07)	1. Collect TTS debug log from all §4 runs. 2. Count words per reply.	Every reply ≤ 15 words.	Not Run
TC-22.2	Edge: long confirmation (NFR-07, EC-05)	1. S1 with C3 mock returning a 25-word `confirmationText`. 2. Play A-01.	Spoken reply is a place-and-minutes summary ≤ 15 words.	Not Run
TC-22.3	No taps while moving (NFR-07)	1. S1 with R1.gpx at 1x (speed > 0). 2. Do each in-trip action by voice only.	Every in-trip action works by voice; none needs a tap.	Not Run

C1 Mic and Mute

ID	Description	Steps	Expected Result	Status

TC-10.1	State shown while listening (REQ-10)	1. S1. 2. Screenshot screen and notification shade every 30 s for 10 min.	All 20 screenshots show “Listening.”	Not Run
TC-10.2	State shown when muted (REQ-10)	1. S1. 2. Tap mute. 3. Check screen and notification.	Both show “Muted.”	Not Run
TC-10.3	Failure: error state (REQ-10)	1. S1 with debug flag `mic_ignore_stop`. 2. Tap mute.	Both show “Error” within 2 s.	Not Run
TC-11.1	Tap mute time (REQ-11)	1. S1. 2. Tap mute. 3. Read time from tap to `micState=muted` in debug log. 4. Repeat 20 times.	≤ 1.0 s in 20 of 20.	Not Run
TC-11.2	Voice mute time (REQ-11)	1. S1. 2. Play A-07 “Veer, stop listening.” 3. Read time from phrase end to muted. 4. Repeat 20 times.	≤ 1.0 s in 20 of 20.	Not Run
TC-11.3	Edge: mute with no network (REQ-11)	1. S1. 2. Set network offline. 3. Do TC-11.1 steps 2-4.	≤ 1.0 s in 20 of 20.	Not Run
TC-12.1	Failure: mic will not stop (REQ-12, EC-04)	1. S1 with `mic_ignore_stop`. 2. Tap mute. 3. Repeat 20 times.	Mic force-stopped ≤ 2.0 s in 20 of 20; “Error” shown.	Not Run
TC-12.2	No audio after mute (REQ-12)	1. TC-12.1 steps 1-2 while playing A-01. 2. Read server frame counter for the trip.	0 frames received after mute time; phone buffer empty.	Not Run
TC-12.3	Edge: mute mid-request (REQ-12)	1. S1. 2. Play A-01 and tap mute 0.5 s in.	In-flight I2 request cancelled; no proposal arrives.	Not Run
TC-13	Unmute by tap only (REQ-13)	1. S1, tap mute. 2. Play A-08 “Veer, start listening.” 3. Tap unmute.	Step 2: still muted. Step 3: “Listening.”	Not Run
TC-24	Background mic and network (NFR-09)	1. S1. 2. Press Home, wait 10 min. 3. Play A-01.	Notification visible the whole time; proposal arrives.	Not Run

C2 Speech Understanding

ID	Description	Steps	Expected Result	Status
TC-02	Confirmation before activation (REQ-02)	1. TC-01.1 steps 1-2. 2. Read TTS debug log.	Confirmation ≤ 15 words is spoken before approval is possible.	Not Run
TC-06.1	Request outside allow-list (REQ-06)	1. S1. 2. Play A-09 “play some jazz.” 3. Repeat 20 times.	`unsupported_constraint`; route unchanged; repeat prompt ≤ 2.0 s in 20 of 20.	Not Run
TC-06.2	Edge: noise only (REQ-06, EC-07)	1. S1. 2. Play A-10 (5 s road noise).	`unclear_speech`; route unchanged; prompt ≤ 2.0 s.	Not Run
TC-06.3	Failure: corrupt audio (REQ-06, EC-07)	1. Run `scripts/send_corrupt_audio.ts` against staging I2.	`invalid_audio` (400); C1 prompts ≤ 2.0 s.	Not Run
TC-16.1	No audio on phone (NFR-01)	1. S1, play A-01 to A-05. 2. Force-stop app mid-utterance. 3. `adb shell find /data /sdcard -name "*.pcm" -o -name "*.wav" -o -name "*.m4a"`.	0 audio files.	Not Run
TC-16.2	No audio on server (NFR-01)	1. After TC-16.1, dump staging DB and logs. 2. Grep for audio bytes and transcript text.	0 matches.	Not Run
TC-16.3	Provider does not keep audio (NFR-01)	1. Open the G-03 written proof in the repo. 2. Check the staging E2 setting matches it.	Proof exists and setting is no-retention.	Not Run
TC-26	Speech accuracy (NFR-11)	1. Play clip set P-01 to P-10 for each of 3 speakers (S1, cabin noise). 2. Compare constraint to answer key.	≥ 8 of 10 correct for each speaker.	Not Run

C4, Auth, Logging, Config

ID	Description	Steps	Expected Result	Status
TC-14	Usage report (REQ-14)	1. Run a trip with 3 AI and 2 Maps calls (1 failed). 2. `GET /api/v1/usage/report` with admin key. 3. Repeat with no key.	Step 2: trip shows 3 AI, 2 Maps, cost > 0. Step 3: 401.	Not Run
TC-18.1	No keys in app (NFR-03)	1. `apktool d veer.apk`. 2. Grep for `AIza` and `sk-`.	0 matches.	Not Run
TC-18.2	Token storage and expiry (NFR-03)	1. Register. 2. Check shared prefs has no token. 3. Set server clock +61 min; call I4. 4. Refresh.	Token only in Keystore; step 3 gets 401; step 4 works.	Not Run
TC-18.3	Edge: register rate limit (NFR-03, EC-17)	1. Send config limit + 1 register calls from one IP in 1 min.	Last call gets 429 `rate_limited`.	Not Run
TC-19.1	Alert at 80% (NFR-04)	1. Set AI test cap $10. 2. Spend to $8.00, then $8.50.	1 alert at $8.00; no second alert (EC-21).	Not Run
TC-19.2	Deny at 100% (NFR-04, EC-13)	1. Spend to $10.00. 2. Make 1 more AI call.	429 `quota`; route unchanged.	Not Run
TC-19.3	Edge: concurrent spend (NFR-04, EC-14)	1. Set used to $9.90. 2. Fire 20 concurrent $0.05 calls.	≤ 2 allowed; `used` ≤ $10.00.	Not Run
TC-20	Log redaction (NFR-05)	1. After all §4 runs, grep staging logs for transcripts, lat/lon, tokens, keys. 2. Parse every line as JSON.	0 matches; every line is JSON with `requestId`.	Not Run

TC-21	Bad config stops startup (NFR-06, EC-27)	1. Delete `timeout.planSec` from config. 2. Start backend.	Startup refused; `config_invalid` logged.	Not Run
TC-23	Retention (NFR-08)	1. Seed events aged 89 and 91 days and totals aged 11 and 13 months. 2. Run retention job.	Only the 91-day events and 13-month totals are deleted.	Not Run

Constraints

ID	Description	Steps	Expected Result	Status
TC-27	Team-funded (CON-01)	1. Check monthly bills for Render, OpenAI, Google, Open-Meteo.	Each is within its cap; app runs on a stock phone.	Not Run
TC-28	No audio; indicator and mute (CON-02)	1. Confirm TC-10, TC-11, TC-16 passed.	All pass.	Not Run
TC-29	Voice-only while moving (CON-03)	1. Confirm TC-22.3 passed.	Pass.	Not Run
TC-30	Google Maps terms (CON-04)	1. Check DB schema for place data. 2. Check the base map.	No place table; Google base map only.	Not Run
TC-31	Short replies (CON-05)	1. Confirm TC-22.1 and TC-22.2 passed.	Both pass.	Not Run
TC-32	Fair ranking (CON-06)	1. S1 with fixture F-SPONSOR (one place marked sponsored, longer detour). 2. Play A-01.	Shorter-detour place chosen; sponsored flag ignored.	Not Run

5. Test Environment

Environment	Used for	Setup
Development	Unit, integration	Laptop; Node.js 20 LTS; PostgreSQL 16 in Docker; Android Studio with Pixel 7 emulator, Android 14 (API 34); E2 to E4 mocked with recorded fixtures. GitHub Actions runs the same on every push.
Staging	System, load	Render web service + managed PostgreSQL 16, separate from production; real E2 to E4 keys with the $10 test caps; Pixel 7 emulator as above.
Pilot	Acceptance, validation	Team Android phone (Android 14 or later), car mount, team member’s car; production build on Render. Parked car at M-20, supervised drives with a non-driving tester at M-23.

Test data (in `test/fixtures/`): audio clips A-01 to A-14 and accuracy set P-01 to P-10 from 3 speakers (16 kHz mono WAV); GPS track R1.gpx; Maps fixtures F-NEAR, F-FAR, F-SPONSOR; weather fixtures W-RAIN, W-STALE; DB seed script `seed_test.sql` (one installation, budget rows at 0%, 79%, 99%). **Network:** emulator set to LTE for normal runs and to offline for TC-11.3 and TC-15.3. **Config:** `config/test.json`, checked in.

6. Pass and Fail Criteria

Per test case. *Pass:* every expected result in §4 is met in every trial, at the §3 threshold. *Fail:* any trial misses any threshold, or the active route changes when it should not. *Blocked:* the case cannot run because of an environment or dependency problem (logged with the reason). *Not Run:* not yet tried.

Defect severity. Sev 1: privacy or safety (audio kept, mute fails, route changes without “okay”). Sev 2: a Must case fails. Sev 3: a Should case fails. Sev 4: cosmetic.

Whole plan. *Pass:* 100% of Must cases pass, at least 90% of Should cases pass, 0 open Sev 1 or Sev 2 defects, and validation meets its §3 target. *Fail:* any of these is missed at M-23.

7. Schedule

Milestones are from the Week 8 plan.

Milestone	Test activity	Owner	Evidence
M-10 Nov 11	This plan submitted	Grady Algire	Test_Plan.md in repo
M-15 Dec 2	Unit tests and CI for config and C4 tables (TC-21, TC-23)	Aiden Ward	Green GitHub Actions run
M-17 Feb 1	Unit + integration: auth and logging (TC-18, TC-20)	Aiden Ward	CI run; test log
M-18 Feb 12	Integration: C2 (TC-02, TC-06, TC-16); first TC-26 run	Brady Cooper	CI run; accuracy sheet
M-19 Feb 15	Integration: C3 (TC-01, TC-05, TC-07, TC-08)	Briar Elliot, Dominic Rowland	CI run; test log
M-20 Mar 3	System in parked car: mute, approval, guidance (TC-03, TC-04, TC-10 to TC-13, TC-15, TC-22, TC-24); first validation sessions	Aiden Ward, Grady Algire	Test log; screen recordings; validation notes
M-21 Mar 15	System: full edge-case suite and load (TC-17, TC-25, TC-26)	Each component owner	Edge-case report; k6 report
M-22 Mar 16	System: usage and caps (TC-14, TC-19)	Aiden Ward	Test log; alert email
M-23 Apr 2	Acceptance: supervised drives and advisor review; constraints (TC-27 to TC-32)	Grady Algire	Drive log; validation results; defect list
M-24 Apr 8	Full regression on the v1.0 tag	Grady Algire	Test results table

8. Definition of Done

Feature done when:

- [ ] Unit tests written and passing in CI
- [ ] Its §4 cases pass, including the edge case
- [ ] Code reviewed by one other team member
- [ ] Docs and config updated (Tech Spec, README if needed)
- [ ] No open Sev 1 or Sev 2 defects on it
- [ ] Component owner signs off in the PR

Project done when:

- [ ] All §4 cases run; whole-plan pass criteria in §6 met
- [ ] Traceability matrix Status column updated for every row
- [ ] All docs match the v1.0 build
- [ ] 0 open Sev 1 or Sev 2 defects; Sev 3 and 4 listed in the final report
- [ ] Team and advisor sign off at M-23